BASC0024

Towards a Literary Machine Translation: An Experiment with French Realist Literature

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May 2020

Abstract

With advancements in machine translation (MT) technologies, the prospect of a machine capable of translating literary works is becoming less farfetched. The aim of this project was to explore how MT could be specialized for literature translation. The experimental approach involved building and adapting (using the cloud-based KantanMT platform) a French-English statistical machine translation (SMT) engine trained on a corpus of French Realist literature novels and their English translations. In preparation for the experiment, essential analysis was made in the field of machine as well as literature translation. The history and technologies of MT were reviewed, as well as the main issues that the field faces in terms of output text quality. In addition, fundamental principles of literature translation were discussed, along with a critical analysis of previous studies on MT for literature.

The detailed methodology of building and training the French Realist SMT engine was described, as well as automatic evaluation of the engine. Finally, a human evaluation survey was sent to a pool of 12 readers to assess the quality, fluency and literary capabilities of the raw output when translating an extract from a French Realist novel. In comparing to the output of Google Translate on the same source text, most readers judged that the French Realist engine, despite being less fluent, produced more valuable literary instances. Although MT is currently not at a level where it is comparable to human literature translation, future technologies may have the capability to increasingly assist the human process.

Preface

At first, the topic of machine translation was appealing to me for the way it combines several interests of mine and aspects of my personality. As a speaker of several different languages I am fascinated by the way that different languages affect not only my self-expression but also the way the world is perceived and categorized. The transfer of ideas and cultures between languages is hardly straightforward or absolute. In this sense, the very concept of machine translation seemed nearly oxymoronic. Being a frequent user of Google Translate for the purposes of language classes I have taken throughout my education, I am acutely aware of the slight imperfections and at times amusing results delivered by a machine that is innocent of the context in which a translation is required. At the same time, I am greatly interested in the technical components which allow a machine to produce even the simplest of translations, an already impressive feat. Having studied computer science and programming, I was interested in learning the workings of a machine translation engine by practically creating one of my own.

The literature component of this project is another interest of mine. Besides being an avid reader, I have pursued in university classes in literature and creative writing. In its creative and elusive quality, literature translation is arguably the ultimate 'Turing Test' for machine translation.

This dissertation project therefore involved several different facets, from the technical review of machine translation, review of the fundamentals of literature translation, and finally the practical and technical skills of producing a custom engine. Machine translation is itself a highly interdisciplinary field involving, for example, computer scientists and linguists. When specializing machine translation for literature, the research involves more levels of complexity, especially in terms of conceptualizing an engine and evaluating the extent to which a translation of literary text is valid and valuable. For example, the traditional automatic evaluation methods of evaluation of machine translation are not entirely appropriate for an engine intended to translate literature. In this respect, this project also involves a small-scale human survey for potential readers, another aspect of interdisciplinary research.

Acknowledgment

Thank you to my family for the support, and to my cousin Carlota for the PhD expertise in writing dissertations.

Lastly, this project would not be possible without the continued help of my supervisor Emmanouela. For the academic help in learning a new area from scratch and for the practical and technical guidance in building a custom machine translation engine, thank you.

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List of Abbreviations

ALPAC – Automatic Language Processing Advisory Committee

BLEU – Bilingual Evaluation Understudy

CEC – Commission of European Communication

EBMT – Example Based Machine Translation

EEC – European Economic Community

FAHQMT – Fully Automatic High-Quality Machine Translation

FRMT – French Realist Machine Translation

GT – Google Translation

MT – Machine Translation

NMT – Neural Machine Translation

SL – Source Language

SMT – Statistical Machine Translation

TER – Translation Error Rate

TL – Target Language

TMX – Translation Memory Exchange

UTF-8 – 8-bit Unicode Transformation Format

1. Introduction

Historically, Machine Translation (MT) has been most successful in the translation of formal texts such as medical or scientific writing (Somers, 2003, p.176). Publishable quality, however, is only achieved through the use of post-editing – the human process of amending or perfecting the raw output.
On the other side of MT are general-purpose engines such as Google Translate. However, despite being intended to be used for any purpose, this type of engine still falters in the translation of more nuanced, creative texts (Quah, 2006, p.66).

The prospect of Fully Automatic High-Quality Machine Translation (FAHQMT) translating Shakespeare or Tolstoy has long been idealized. Unlike technical manuals or reports, literary works are complicated and cryptic, involving creativity and subtlety to express meaning in unique ways. This nuance is a source of problems for MT, as language in literature cannot be categorized and predicted the same way that it is for scientific reports or legal documentation. For this reason, MT-for-literature researchers such as Toral and Way (2018, p.2) regret that there is "no prospect of machines being useful at (assisting with) the translation of this type of content". Indeed, for a machine to deterministically learn the art of literature translation – that is itself opinionated and undefined – is a tremendous task. Still, this project aims to explore the properties of MT that can yet develop to better support MT for literature. With technological advancements in MT architectures and performances as well as increased open access to full literary texts on the web, today is a time of experimentation for MT in literature.

The purpose of this project is firstly to provide an overview of the field of MT as well as of literature translation, analyzing the ways in which the two fields intersect in order to further the research in MT for literature.
Ultimately, an experiment is devised to assess whether there is an argument for MT applications in literature translation. A French-English statistical machine translation (SMT) engine is built with the intention of specializing it for a specific literary genre, 19th Century

French Realist literature. The resulting engine is critically assessed through human evaluation of the system's translation of a literary text from the specified genre, also comparing it to the output produced by Google Translate on the same text.

In this research that re-defines itself as new technologies and approaches become available, incremental (however small) improvements and findings are the desirable way forward.

2. Machine Translation Literature Review

2.1. Brief History of Machine Translation

There are many claims regarding the origins of the concept of the mechanization or automation of translation, with enthusiasts such as DuPont (2018) stretching as far back as the cryptanalytic translation procedures of Arabic scholars around the 13th century. In a more conservative view, Hutchins (1992) traces Machine Translation (MT) concepts back to the 17th century, where Descartes and Leibniz suggested the use of mechanical dictionaries (based on universal numerical codes) to overcome language barriers. However, when strictly considering the machine component, the birth of MT is traditionally attributed to the American scientist and mathematician Warren Weaver, who wrote the pivotal 'Translation Memorandum' in 1949 (Arnold et. al, 1994). Indeed, Weaver does conceptualize MT as a matter of cryptology; de-coding a given language and restructuring it with the rules of another. The following is a section of Weaver's Memorandum (Arnold et. al, 1994):

"I have a text in front of me which is written in Russian but I am going to pretend that it is really written in English and that it has been coded in some strange symbols. All I need to do is strip off the code in order to retrieve the information contained in the text."

The memorandum – which included stricter and more objective language than the above – proved to be greatly influential, as it triggered a surge in interest and investment in MT. Throughout the 1950s, research spiked in Europe and the USA, with one of the first demonstrations of a working albeit minimal MT system in Georgetown University in 1954 (Hutchins, 1992). The system (which translated Russian sentences into English) was limited to a vocabulary of 250 words and employed a total of six grammar rules. Despite the minimal scientific findings, the demonstration motivated continued and increasing funds for MT, and interest was sparked elsewhere, notably in the Soviet Union (Hutchins, 1992). Much of the investment and research in MT stemmed from Cold War tensions, in order to gain intelligence on the enemy.

The emergence of MT, in being complex and timely, brought along with it a number of skeptics, who doubted its theoretical and economic feasibility. In 1964, the funding authorities for MT decreed the formation of the Automatic Language Processing Advisory Committee (ALPAC), who were to report on the current advances, successes, and failures in MT research. In the important 1966 ALPAC Report, the committee concluded that MT was unlikely to produce any useful translations in the near future. Moreover, the report judged that MT was too expensive, and that there was no shortage of human translators (Arnold et. al, 1994). The report was traumatic for MT research, and halted government funding in the USA. To this day, many of the arguments of the ALPAC Report are still entertained by MT skeptics.

As a result of the ALPAC Report, most MT research took place outside of the USA during the 1970s. Research continued in Canada, where the METEO system was developed by the TUAM group of the University of Montreal (an English-French translator) and Western Europe, who required translations of technical, scientific, and legal documentation from and to every of the language of EEC (European Economic Community) members (Hutchins, 1992). The Commission of European Communications (CEC) invested in an English-French version the SYSTRANS system, an improved version of the early Georgetown University MT system. Moreover, the CEC commissioned French-English and Italian-English versions of SYSTRANS. In addition, in the late 1970s, the EUROTRA project was established, which would become one of the largest Natural Language Processing projects (Arnold et. al, 1994).

The 1980s and 1990s consisted largely in the development of the projects and systems mentioned previously. Up until the early 1990s, MT systems were rule-based, modeled from linguistic rules specific to each language. However, the 1990s saw the arrival and rise of data-driven methods for MT, most importantly statistical machine translation. It was IBM who spearheaded the development of statistical machine translation (SMT), with Google close behind. More recently, the surge in interest in artificial intelligence and deep learning has sprung the development of neural MT systems (Jones, 2017). With large amounts of data and strong computing power at hand, Language Service Providers such as KantanMT (the software used in this project), have the ability to offer customizable MT engines designed for specific domains.

2.2. Types of MT Systems

The three main types of MT systems are – in historical order – direct systems, rule-based systems, and corpus-based systems (Quah, 2006). Corpus-based approaches have proven, in recent years, to be the most effective of the strategies. However, it is worth tracing the different approaches and advancements made since the beginnings of MT.

The **direct** approach, used in the first forms of MT, involved the use of bilingual dictionaries in order to match Source Language (SL) words to their equivalent in the Target Language (TL) (Quah, 2006). The design follows the word-for-word paradigm in translation, and the direct, rudimentary nature of these systems involves a small morphological analysis, and altogether lacks any lexical or semantic analysis of either the SL or TL (Hutchins, 1992).

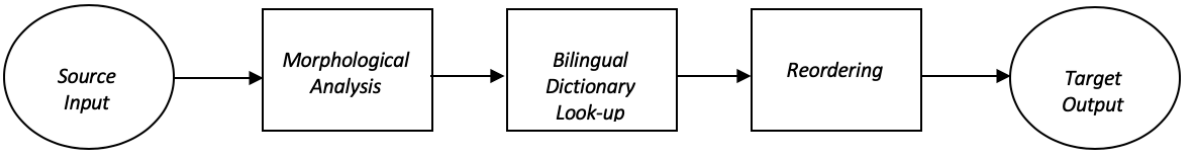

Fig. 1: Direct Approach. Adapted from Hutchins (1992, p.72).

The onset of the **rule-based** approach was largely a response to the lack of significant language analysis in previous methods (Hutchins, 1992). The system analyses the source text by using morphological, syntactic, and semantic rules, and produces target output by using rules specific to the TL (Quah, 2006). The two types of architectures associated to rule-based systems are interlingua and transfer systems. In interlingua systems, languages are exhaustively analyzed in order to synthesize an 'intermediate' language which can in turn be used for the synthesis of a given target language. The idea is that an abstract, universal language can be used to represent both the SL and the TL (Hutchins, 1992). This paradign simplified the process of adding language combinations, involving the analysis and blueprinting of a language in relation to the intermediate, neutral language.

Fig. 2: Interlingua Approach. Adapted from Hutchins (1992, p.74).

The transfer architecture is less abstract than the interlingua, as it uses language specific intermediate stages rather than an all-encompassing neutral language. The SL is parsed using specific grammar rules and is then transferred to a representation of the TL. Finally, natural TL text is synthesized from the TL representation (Isabelle and Foster, 2006).

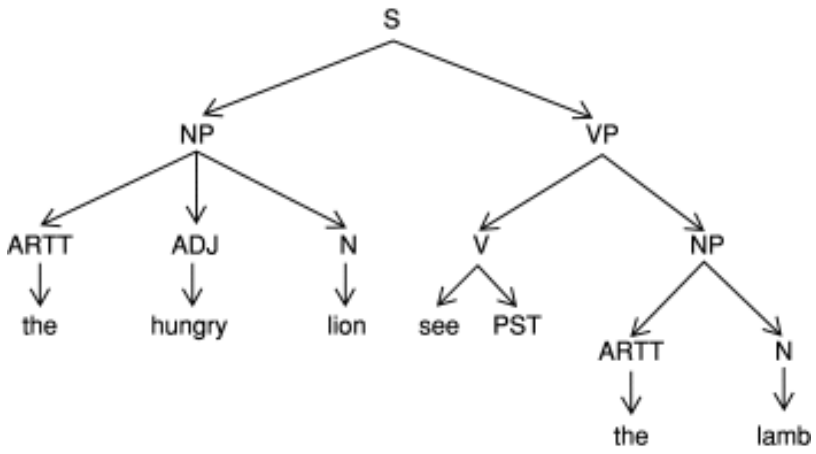

Fig. 3: Transfer approach – English representation. Adapted from Isabelle and Foster (2006).

The most recent approach to MT design is the **corpus-based** approach, which involves three different architectures: example-based, neural MT, and most crucially in the context of this project, statistical MT. These systems make use of a large corpus of monolingual or bilingual texts in order to generate new translations (Patiniotaki, 2020).

The idea behind **example-based** MT (EBMT) is to generate new translations by comparing sentence segments to existing translations inside the corpus using matching algorithms (Quah, 2006). For example, to translate sentence A, the system will match with existing translations B and C as shown:

(A) I organized my summer vacation.
(B) I organized my studies. → J'ai organisé mes études.
(C) She went on a summer vacation. →Elle est partie en vacances d'été.

The system then reorganizes the matched segments to produce the required translation.

(D) I organized my summer vacation. → J'ai organisé mes vacances d'été.

The EBMT system will match with progressively smaller sections of existing translations, eventually down to specific words. The more translations the system needs to match, the more complex and erroneous the translation is likely to become (Isabelle and Foster, 2006).

The most popular type of corpus-based approach is **statistical** MT (SMT). Importantly, the KantanMT system used for this project is a SMT engine, therefore a separate section is be dedicated to the subject. Broadly speaking, SMT introduces the application of statistical methods to calculate the probabilities of new translations using prior knowledge found inside the bilingual corpus (Quan, 2006). Given an input of a SL segment, the engine estimates the probability of the TL output by performing parameter matching algorithms and running them through a probabilistic model. The mathematical models are centered around Bayes theorem, which assesses the probability of a translation given a certain input (further details in Section 2.3 on SMT).

Neural MT (NMT) is the most recent development in corpus-based approaches, using machine learning algorithms of neural networks. As shown in Figure 4, the engine processes a vector of input nodes (SL text) through *n* hidden layers before generating a vector of output nodes (TL text).

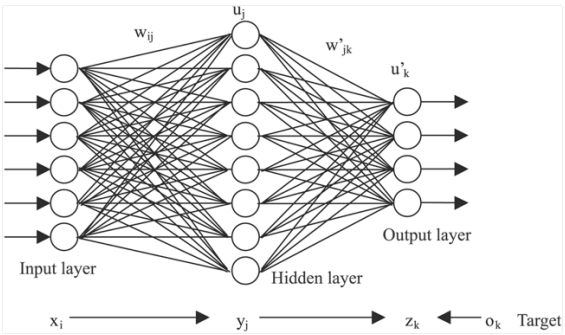

Fig. 4: Representation of a Neural Network. Adapted from Koehn (2017, p.8).

A neural MT is trained on a large aligned bilingual corpus (each SL input is paired with a corresponding TL output), where every component (weights and bias between layers) is adjusted to generate successful output (Cho, 2015). Although after the training process a neural network is technically deterministic, they remain a 'black box' algorithm in that the internal workings of the hidden layers are not visible and the effect of training is unpredictable (Koehn, 2017). Besides offering the promise of increased accuracy in translations, neural MT in also significant in promoting the field in terms of available, reproducible, and customizable solutions in MT (Jones, 2017).

2.3. Statistical MT

As mentioned previously, a SMT engine contains two key components, a translation model and a language model, as shown in Figure 5.

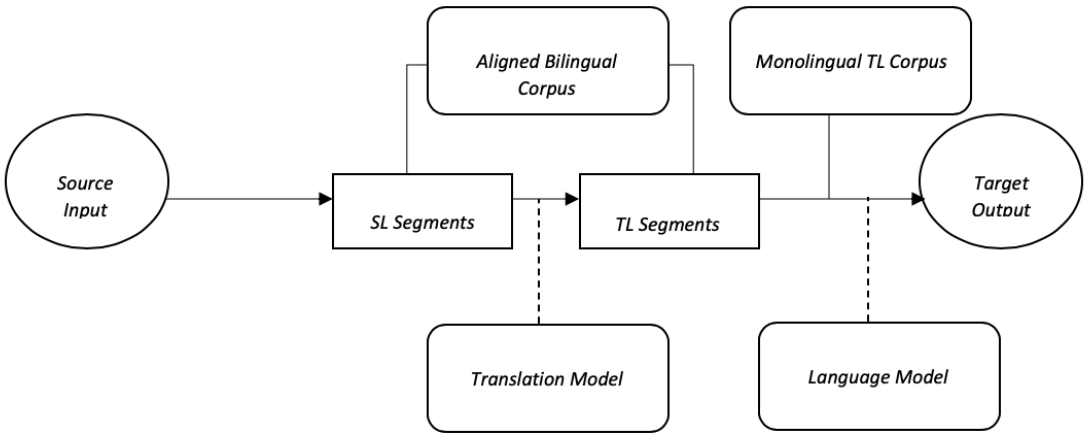

Fig. 5: SMT System Diagram. Adapted from Quah (2006, p.78)

The translation model is extracted from the aligned bilingual corpus, which contains SL segments linked to their equivalent in the TL. A segment has a variable length, from a single word to several sentences, and each SL segment is associated to a single TL segment (Isabelle and Foster, 2006). However, given that words and phrases are repeated throughout the

parallel corpus, a SL word/phrase will often correspond to more than one translation in the TL. The principle of the translation model is to compute the probability of a translation being equivalent (i.e. faithful) based on the occurrences of substrings in the corpus. The size of the corpus as well as the quality of the alignment are determining factors in the quality of the translation model, and in turn the SMT engine (Hearne and Way, 2011). As mentioned in Section 2.2, SMT is based on Bayes Theorem of Probability, and each of its components plays a specific role in the calculation of a best-estimate translation, $P(T|S)$.

$$P(T|S) = \frac{P(T) \times P(S|T)}{P(S)}$$

Fig. 6: Bayes Theorem Equation.

Mathematically speaking, the objective an SMT engine is to maximize the numerator, $P(T) \times P(S|T)$. The translation model of the engine is focused on calculating the value of $P(S|T)$, which is the probability that "given target candidate *T*, source sentence *S* is likely to have given rise to *T*" (Hearne and Way, 2011). Naturally, if the *S*-*T* pair is similar to translations found within the corpus, the value of $P(S|T)$ will be higher.

Alongside the translation model, the system combines information from an associated language model to estimate the most adequate translation (Quan, 2006). The language model is extracted from a monolingual corpus of the TL, through which morphological, grammatical and linguistic information is processed. Earlier versions of SMT developed by IBM work on a basis of word-to-word alignments, where a SL word aligns to one target word, whereas TL words can have any number of SL word connections (Isabelle and Foster, 2006). This approach invariably leads to structural inaccuracy in the form of word order in target language, a phenomenon known as 'distortion'. In generating TL text, statistical methods are employed to predict the correct word order. For example, $P(4|5)$ assesses the probability with which source word in position 4 will generate a target word in position 5 (Quan, 2006, p.79). More recent versions of SMT also use phrase-based alignments, which account for greater ambiguities in natural language that extend beyond words and structure.

The following example is a simple illustration in translating a single word in a French-English engine. Assuming the word *grand* has several possible translations in an example parallel corpus (see Table 1), the specific probability of each can be determined.

Source word/segment [occurrences in parallel corpus]	**Candidate translation [occurrences in parallel corpus]**	**Candidate translation [occurrences in parallel corpus]**	**Candidate translation // occurrences in parallel corpus**
'grand' [15,000]	*'big'* [3,000]	*'large'* [4,000]	*'great'* [5,500]

Table 1: Different translations of SL word 'grand' and their occurrences in example corpus.

Intuitively, the most likely translation is *great*, and indeed the translation model identifies the substring *grand* as translated most probably to *great*. In such a simple case, a translation model is sufficient, but certainly when full sentences of the SL are input, the language model is essential. Given a candidate target sentence output (deduced by the translation model), the language model's purpose is to construct appropriate and fluent text by computing the possible word combinations. In doing so, the probabilities are split amongst the constituting words, as shown in Figure 7.

$$P(T) = P(T_1, T_2, T_3, \dots, T_i) \ \ (where\ T_i\ is\ a\ word\ in\ position\ i)$$

Fig. 7: Probabilistic constructing TL sentence.

For the sentence '*the mountain is […]*', that probability of getting *great* is calculated using the formula for conditional probability, where P(A | B) = P(A ∩ B) / P(B).

Candidate translation [occurrences in monolingual corpus]	**Candidate translation [occurrences in monolingual corpus]**	**Candidate translation [occurrences in monolingual corpus]**	**Candidate translation [occurrences in monolingual corpus]**
'the mountain is' [10,000]	*'the mountain is big'* [2,500]	*'the mountain is large'* [1,000]	*'the mountain is great'* [2,000]

Table 2: Candidate translations and their occurrences in example monolingual corpus.

$$P(great|the\ mountain\ is) = \frac{P(the, mountain, is, great)}{P(the, mountain, is)}$$

$$P(great|the, mountain, is) = \frac{2{,}000}{10{,}000} = 0.2$$

Fig. 8: Conditional probability for 'great' given sequence of words.

This rudimentary example of probabilities demonstrates some basic operations occurring in the translation and language models of an SMT engine. Importantly, they demonstrate how influential the quality and size of the corpora are for the successful operation of the engine (Quah, 2006), as it influences the engine to make certain word/phrase choices over others.

2.4. Main Problems in MT

Many of the problems that the field of MT currently faces in terms of output quality have been ongoing since the very inception of MT. According to Arnold et. al. (1994), ambiguity is the main source of problems for MT systems, and exists in different forms. Words that carry more than one meaning are lexically ambiguous, whilst phrases or sentences that can take on several meanings are called structurally ambiguous. Lexical ambiguities typically exist in three different forms: category ambiguity, synonyms and transfer ambiguity (Schwarzl, 2001, p.34). Mismatches between the formulation of different languages is another source of errors for MT. Finally, there are idiomatic elements which can cause confusion given their different structural and semantic meaning.

2.4.1. Lexical Ambiguities

i) Category Ambiguity

Category ambiguity involves words that belong to more than one grammatical category, the most common of which are noun-verb pairs.

French: *lit* → English: *bed (noun), read (verb)*

These types of ambiguities are dealt with by the syntactic and morphological rules of the target language, but if a sentence contains several category ambiguities the problem becomes exponentially more intricate.

ii) Synonyms

A word that can have different translations causes problems for MT, whether the synonymity is subtle or semantically pronounced. Corpus based systems intrinsically teach the engine the context of each word. In general, systems deal with this issue by associating semantic tags to a given lexical entry (Schwarzl, 2001).

iii) Transfer Ambiguity

Whereas category ambiguity and synonyms happen at the analysis and parsing stage, transfer ambiguity relates to the generation of target language. A word which is considered unambiguous in the SL but translates to an ambiguous TL word is a transfer ambiguity, also called a translation ambiguity.

French: *piquet* → English: *picket, stake*

A *piquet* is either a *stake* or a *picket*, depending on whether it has been driven into the ground. These issues are specific to the language pair and require further syntactic as well as semantic rules.

2.4.2. Structural Ambiguities

Structural ambiguities occur when there is more than one possibility of structuring a string of words. One common example is the sentence *flying planes can be dangerous*, which can be understood in two different ways. Such phrases or sentences can produce erroneous parsing by MT systems and therefore compromise the output. Another specific type of structural ambiguity are cases of attachment ambiguity. These occur when it is unclear where prepositional phrase or relative clause attaches to. The following example is an illustration of this phenomenon (Schwarzl, 2001): *The man saw the horse with the telescope.* For a human reader the sentence is rather obvious, but syntactically the sentence can also describe a man seeing a horse who possesses a telescope. Another example of structural ambiguity happens

in dialogue translation, where each phrase of is interwoven with the previous dialogues. Nuanced aspects like the position and tone of the speaker compared with the listener are hard to grasp without context and previous knowledge (Arnold et. al, 1994). In most of these cases, real-world knowledge is required to correctly analyze the structure of sentences.

2.4.3. Lexical and Structural Mismatches

The way in which a given language reads and classifies the world is always unique. As a result of fundamental cultural specificities, some languages may deliver a message with a given structure to deliver a message in the way it is delivered in another language. Some structures may be equal, but used in different ways, and some structures may be different but used for the same objective. This is particularly true for cultures with greater diversions. There is a greater mismatch between English and Japanese than there is between English and French (Arnold et al. 1994).

These are the issues that human translators deal with, and arguably are the reason why there is such a thing as a good and bad translator. The cultural specificities attached to languages are hard to abstract, and the job of a translator lies in the decision of how to do so. In this largely creative aspect of translation, computers unequivocally pale in comparison to human translators.

2.4.4. Idiomatic Elements

Idioms can pose issues to MT because engines are innocent to the (more often than not) pronounced disparity between literal and intended meaning. We can consider the following French idiom:

French: *Coûter les yeux de le tête.*
Literal Translation: *To cost the eyes off the head.*
English equivalent: *To cost an arm and a leg.*

The literal translation and the English equivalent are thematically similar, but to use one over the other has vast semantic and cultural implications. Idioms defy logic in language, introducing complexity and nuance not only at a semantic level but also influencing resulting syntax and morphology. With corpus-based approaches, systems can learn the correct

translations of idioms given that they appear sufficiently inside the bilingual corpora (Anastasiou, 2010). However, further complications can occur if language plays on idioms themselves, where an authorial intention must be interpreted correctly.

Naturally, language persistently works in all of the ways previously described, and oftentimes combines a variety of issues. For example, a sentence can contain several ambiguous words and involve one or more idioms, while itself being structurally ambiguous. Far from being a one-to-one function, translation is a many-to-many interaction between source and target, whereby an MT engine must select a unique output.

Finally, an exercise with Google Translate is enough to demonstrate that the prospect of a universal, general purpose FAHQMT is still unrealistic. Figure 9 shows three different translations of an excerpt of Saint-Exupery's *The Little Prince*.

French (Original)	
Je demande pardon aux enfants d'avoir dédié ce livre à une grande personne. J'ai une excuse sérieuse : cette grande personne est le meilleur ami que j'ai au monde.	
Google Translate	**Katherine Woods Translation**
I apologize to the children for having dedicated this book to a tall person. I have a serious excuse: this grown-up is my best friend in the world.	*I ask the indulgence of the children who may read this book for dedicating it to a grown-up. I have a serious reason: he is the best friend I have in the world.*

Fig. 9: Comparing Google translate to human translation.

In this case, Google Translate is (although thematically fitting) innocent to the intended meaning of a *grande personne*, translating as *tall person*, although in the second instance the engine translates correctly as *grown-up*. In this case, authorial intention is paramount, and adds more layers of nuance, including authorial voice and style. It is worth noting that there are numerous different human translations of the above sentences, all of which have been accepted and published. Apart from the aforementioned instance, the Google translation is certainly a legitimate one, a hopeful prospect for the field.

3. Literature Translation and MT for Literature

For the scope of this project – as discussed in the introduction – the applications of MT for literature are considered in the context of and in preparation for an experiment with a French-English MT engine trained on a bilingual corpus of 19th Century French Realist literature. The discussion on literature translation is therefore tied specifically to the French-English language pair, with occasional discussion of the Realist genre. Moreover, the previous studies in MT for literature are predominantly in the appropriate language pair.

3.1. Literature Translation

Before delving into the applications of MT in literature, it is important to understand the questions and processes relating to the field of literature translation itself. The translation of literature is arguably the most culturally significant application of translation, and certainly the most historically contentious (Newark, 1989). Despite the cohesive interest in the global communication of cultures and ideas, each translation of a piece of literature is unique in its own right, and follows to varying extents the procedures and guidelines outlined by translation theories (Newark, 1989). Arguably, the most persistent historical debate within the field is the literal versus the free translation (Venuti, 2012). Whereas literal translation is aimed at a faithful (ideally identical) reconstruction of the original work, free translation is concerned with the transfer of culture and meaning between languages. Regardless of the strategy, all translators seek to produce a quality translation, which lends to the fundamental question: what makes a quality translation?

Venuti (2012, p.5) outlines three pillars to the translation of literature as equivalence, autonomy, and function. Although there exist a multitude of translation theories, these three aspects are applicable to the assessment of any literary translation.

3.1.1. Equivalence

Equivalence is arguably the most important albeit challenging aspect of any translation. Essentially, a translated text aims to be 'equivalent' to the original at several different levels; word, grammar, structure, and thematic (Baker, 2011, p.4). In literature, each word or sentence is a motivated choice by the author, and it is the translator's job to replicate these choices (House, 2014). In practice, however, the specific tools for cultural expression are unique to each language, and the target language will oftentimes be devoid of the grammatical means of expression which exist in the source language (Venuti, 2012). For example, the French word *retrouvailles* – which describes an emotional meeting between individuals after an extended period of absence – is best translated to English as *reunion*, but this translation lacks in the emotional component. Moreover, mismatch between languages can occur on larger and more abstract levels. A notable example of this is the different gendering of language in French and English. Whereas the French language is gendered (nouns are either feminine or masculine), English does not employ grammatical gender. Another aspect of French which is lacking in English is the use of pronouns, where people may be addressed familiarly as *tu* or courteously as *vous*. Translating this important nuance into English is difficult, since the use of the pronoun *you* works ambiguously (Anderman, 2010). In the case of Realist literature – which is centered around themes of high society, marriage and wealth – novels contain significant amounts of dialogue. Naturally, the pronouns of address (in French) provide insight into the relationship between characters. Given the lack of grammatical and cultural parity between languages, translators must use the available means of the target language to maximize the equivalence between the two.

Equivalence above word level, where collocations or idioms are concerned, is also of great importance to the translator. For example, the French collocation *gagner du temps* has the direct translation *saving time*. Idioms, which are similar to collocations but more intrinsically linked to history and culture (Baker, 2011) can be the source of more problems for translators. Firstly, a translator must be able to identify that they are dealing with an idiomatic expression, and then consider if and how an idiomatic expression may be used in the target language. The equivalence of an idiom is necessarily harder to achieve than that of a word, and occasionally misleading. For instance, the French *donner sa langue au chat* ('give one's tongue to the cat') is associated with the giving up when asked a question or riddle. The apparently obvious translation in English, *cat got your tongue?*, is used in the context of urging somebody

to answer when their silence becomes a nuisance (Baker, 2011). A more appropriate translation to English might be *to throw in the towel*. It is up to the translator to interpret and mediate any difference that may result from translation.

In all cases, a translator must manage the equivalence of the text and its translation, and whether the objective of equivalence in one instance may detract from other aspects of the original.

3.1.2. Function

Function, the second pillar of Venuti's (2012) literature translation, relates to a translated work's ability to produce a similar effect to that which the original text has produced in its own culture (Venuti, 2012).

Unlike equivalence, which references the text's connection to the translation specifically in terms of language, function is concerned with the messages and meanings that are interwoven within the work. This is arguably the reason for the existence of numerous translations for a given work, as the effect on a culture is achieved differently depending on the point in history as well as the translator's perception of readership (House, 2014). Realist literature, for instance, is intimately tied into the society and locales it describes. For example, Flaubert's *Madame Bovary* – which follows a woman's apathy towards her provincial life and her will to escape it – is specific to French society and the relationship between the elite and provincial. Culturally, the theme does (to some extent) have an English equivalent; for example Charlotte Brontë's *Jane Eyre*, published in the same period as Flaubert's masterpiece. As such, a translator who privileges function would place great importance on the English reader (with cultural and historical baggage) of *Madame Bovary*. Certainly, the focus on the transferal of the intended messages of a literary work will often interfere with the intention of maintaining a high equivalence. The translator therefore has several jobs: firstly, she is an appreciator and interpreter of literature who puts herself in the skin of the author with the intention of understanding the motivated choices that are made for any particular textual as well as thematic element. Secondly, the translator must turn towards her own culture (or the target culture) imagining and simulating of such choices and ideas in order to transmit them analogously to the said culture (House, 2014; Cronin, 2012). In a sense, the translator is very much an author in her own right, and must

possess the qualities of a writer in terms of writing with quality, clarity and delicacy, all whilst portraying larger, sub-textual meaning (Newark, 1989).

3.1.3. Autonomy

In translation of literature, autonomy is a measure of its success as a literary work. Venuti (2012) defines autonomy of translation as "the factors that distinguish it from the source text and from texts initially written in the translating language" (Venuti, 2012, p.4). In other words, it can be described as the merit that the translating agents (the translator, editor, publisher etc.) have in producing literature that is in itself – without being a comparison of its original – a valuable contribution to a given language, culture and history. Certainly, autonomy is also tightly bound to the balance between equivalence and function, and further highlights the complexity of decisions and motivations that influence the process of translation.

Although there is a wide variety of theories and concepts surrounding the translation of literature, all of these arguably fit into the balancing between equivalence, function, and autonomy (Venuti, 2012). The points discussed above are by no means an exhaustive list of the considerations to be taken in translating literature, but they begin to give an idea as to the complexity surrounding the topic. A translator is faced with an aggregation of linguistic, cultural, historic, and logistical issues; all of which determine that each translation is unique and arguably, personal (House, 2014; Venuti, 2012). Literature, being a creative and artistic medium, requires no less than a creative translator (Newark, 1989). It is in this context that we turn towards the application of MT in literature. Fundamentally, having an MT translate literature is an attempt in simulating the intricate decisions which are involved.

3.2. Applications of MT in Literature Translation

3.2.1. Past Experiments in Literature Translation with MT

The ambition of using MT engines to translate literature is arguably as old as the field of MT itself. As early as 1955, researching pioneers such as Locke and Booth (1955) expressed the ambitions of an MT capable of translating literature. With the present advances in corpus-based approaches (notably SMT and NMT), machines incorporate the patterns of translations found in their training data, which theoretically enables a greater capacity to handle more complex textual entities (Quah, 2006; Isabelle and Foster, 2006).

Another phenomenon that has enabled the research and growing interest in literary MT is the emergence of e-books and online libraries, which allow a more streamlined process in assembling a corpus of texts from the literary domain. However, despite the progress, the present state of MT is far from the level of a FAHQMT for literary text, as several pilot studies show.

In a pilot study in 2013, Jones and Irvine (2013) trained a French-English SMT engine on a parallel corpus from Canadian government proceedings. The conclusions taken from the engine – which was tested on translating prose – highlight the issues and potential successes of MT in literature. Firstly, the study highlights the difficulty of training an MT with the complex processes of translation. As discussed in the previous section, a translator is constantly making decisions regarding the faithfulness to a text and how it may be adapted to the target language. These decisions, which are rarely linear and consistent with one another (considering the different works and translators in a corpus) produce confusion in the MT. The ability of human translators of identifying and managing exceptions in translations is absent in MT counterparts (Jones and Irvine, 2013).

Besacier (2014) performed a similar study, building an English-French SMT engine trained on a parallel corpus from a variety of domains. The engine was tested on an American short story and assessed by a panel of readers. The consensus among the panel was that the MT produced language that – although understandable – did not truly sound like the target language. The idiomatic language and cultural knowledge did not read naturally or fluently. For example, *river Charles* (a location in Massachusetts) was translated to *rivière Charles*,

whereas French topography would typically call it a *fleuve* (denoting a larger body of water than a *rivière*). These instances where specific cultural knowledge is required are a significant source of difficulty for MT (Besacier, 2014).

A more recent study by Moorkens et. al. (2018) asked a group of professional translators to translate using SMT and NMT output (combined with their own post-editing) and compare it to traditional translation from scratch. Both the SMT and NMT engines (English-Catalan) were trained on a corpus of 133 parallel novels, as well as monolingual Catalan data from around 1,000 books. For all participants, post-editing NMT took the least time out of the three methods, requiring the least keystrokes. NMT was more fluent and accurate, but unless limited by strict time constraints, the translators would not choose it over human translation. There was also the opinion that MT can restrict the translator to a solution that is not optimal. On the other hand, some participants argued that MT can give inspiration – or an often necessary kick-start – to translating a given segment (Moorkens et. al., 2018). The idea of a draft translation is arguably the most significant impact that MT currently has in literary translation. However, an empty text box also gives that translator the freedom to exercise the necessary choices and highlight the intimate meaning of the source text, whereas MT often translates too close to the source, leaving a sense that MT conditions a translator towards making a literary translation. Importantly, the translators which took part in the study were critical of the segment-by-segment architecture of MT. Whereas a translator holds a global view of a text, the fragmented view of MT can result in the loss of context and meaning in a piece of literature (Moorkens et. al., 2018).

A subsequent study by Toral and Way (2018) continued the comparison between SMT and NMT (also English-Catalan) by asking native TL speakers to evaluate the machine translations of famous novels from the 1920s to the present day. The study highlighted the greater computational capacity that NMT shows in handling more complex textual features (Toral and Way, 2018). Indeed, for one of the translated novels, one third of participants perceived the NMT translation to be equivalent in quality to that of a human. This increasing acceptance of MT – as well as the promising research in NMT which is still in its early phases – allows for optimism in the pursuit of a literary MT.

3.2.2. Issues with Corpora Selection

All literary MT experiments discussed above are susceptible to a potentially determining factor in achieving results, which is the corpora used for the training of each engine. For example, Jones and Irvine (2013) trained their engine (which translated literary texts) on Canadian Parliament proceedings, while Besacier (2014) assembled a corpus from different domains. This is an issue because for corpus-based methods like SMT and NMT, the resulting machine is only ever as good as the training data provided (Quah, 2006; Koehn, 2010).

MT is strongest in the translation of technical language, and engines are trained on large in-domain corpora to operate optimally (Quah, 2006; Arnold et. al., 1994; Koehn, 2010). As such, the engines produced by Jones and Irvine and Besacier can hardly be identified as specializing in literature. However, even for engines trained on literary text like those of Mookerns et. al. (2018) and Toral and Way (2018), the domain remains arguably too broad. By its very nature, the use of language in literature varies greatly between genres and history. Moreover, a parallel corpus of novels is likely to contain a multitude of different translators, with each employing their own approaches and techniques. In this way, a corpus that spans one century of literature is relatively unspecific, and can generate some confusion in the training process of the engine (Jones and Irvine, 2013). At the same time, the minimum amount of training data remains significantly high (roughly above 2 million words), which results in a trade-off between the specificity and the size of a literary corpus.

3.2.3. Further Issues in Literature Translation with MT

The eventual usage of MT for the purposes of translating literature is not without its broader and more ethical implications. As output quality improves with future research, MT will have increasing implications for authors, translators, readers and publishers. Firstly, there is a sense among the translation community that literature translators are the last ones standing in the strictly human component; and having so far been immune to the surge of MT, their livelihood may be under threat (Taivalkoski-Shilov, 2018). There is extensive debate surrounding this theme, with many arguing that literary translators' expertise will remain essential to the field, and are only likely to be assisted by machines and not entirely replaced by them. On the other hand, the increasing impact of MT can cause a decrease in the compensation given to human translators, in turn affecting the profession and industry (Taivalkoski-Shilov, 2018).

Secondly, considering current issues in MT performance, a low quality of translation of literary works is – according to intellectual property rights – potentially prejudicial to the author's honor and reputation. Besides defamation, poor translations can sabotage an author's work in the target language, eliminating current and potential readers (Taivalkoski-Shilov, 2018). It is paramount that authors should maintain textual ownership of their works – and considering the technical aspects discussed previously, MT poses a potential threat. For readers too, a low-quality translation hinders the transfer of ideas and culture. Moreover, poor grammar and unidiomatic language may even be harmful for readers, if one considers, for example, children's books.

Despite these ethical issues, there is no lack of arguments to defend the ideal of literary MT. Although it is yet to be meaningfully integrated in the literature translator's workflow (Wilks, 2009; Schwarzl, 2001), there is room for growing interest from authors, readers and publishers. As it has proven to speed up the process of translation (Moorkens et. al., 2018), MT is attractive to publishers as it can reduce costs and increase productivity, providing readers' favored authors more readily. For authors, the relatively fast reproducibility of MT allows for the possibility of acquiring more readers from languages that might not have been attainable with the cost and logistics of human translators (Taivalkoski-Shilov, 2018).

Finally, Lee (2011) argues that there is a place for MT where it does not interfere or replace translators' work, but adds something entirely new to the field of literature. In this avant-garde view of MT, the decreased quality is a way of deconstructing a source text and rendering the language borders that exist for a given translation. The formulation of new meanings (whether faithful to the source or not) is in itself a contribution to the understanding of a work of literature (Lee, 2011). Certainly, this rather post-modernist appreciation of MT is not practical in the sense of achieving the objective of a faithful translation. Nevertheless, it demonstrates the blurry state which MT currently occupies with regards to literature, where it is neither an entirely satisfactory tool for the matter, but certainly not a futile endeavor given its insights into translation of language and style.

3.2.4. Observations for Building a Literary MT Engine

As demonstrated by recent experiments in MT for literature, the task remains very ambitious, challenging and complex. This branch of MT is currently in an investigative and experimentative phase. Naturally, in building an MT trained on French Realist literature, one

cannot expect a product capable of competing with professional human translators – it too is an experiment. The particularity of this experiment, compared to the previous studies, is that a literary engine will be built on a more constrained and specific corpus, French Realist literature. In going through the steps of building an engine, the objective is to contribute in some way to the advancing challenge that is translating literature with MT.

4. Methodology

4.1. Preparing the SMT

4.1.1. The KantanMT Platform

KantanMT is a cloud-based web application for building custom MT engines (either SMT or NMT) in over 760 language pairs (KantanMTblog.com, 2020). The streamlined processes of training engines as well as a variety of analytics provided make KantanMT an ideal MT solution for clients both corporate and private. KantanMT also provides numerous domain-specific free training sets for optimizing engines (called KantanLibraries) which were used for the project. (See Appendix I for platform interface.)

4.1.2. Selection of Corpora

The bilingual corpus was assembled from a collection of works from the French literary Realism period, which is generally understood to have begun with the works of Stendhal and Honoré de Balzac in the first half of the 19th Century (Dambrine, 2018). The corpus is restricted to Stendhal, Balzac, and another two important novelists of the Realist period, Gustave Flaubert and Guy de Maupassant. The works range from 1830 to 1887. All French books were found in the Bibliothèque Electronique du Québec and all English translations in Project Gutenberg (See Appendix II for more information). Table 3 shows the details of each novel selected along with the number of words in both SL and TL. Word counts were taken before cleaning the text files, and account for – among longer text segments – formatting differences, such as page numbers in the English versions counting as words.

Author	Original Title	Year Published	# Words in SL (French)	# Words in TL (English)
Stendhal	*Le Rouge et Le Noir*	1830	174,952	191,763
Stendhal	*La Charteuse de Parme*	1839	187,027	196,253
Gustave Flaubert	*Madame Bovary*	1857	115,007	116,626
Gustave Flaubert	*Education Sentimentale*	1869	158,423	162,963
Honoré de Balzac	*Le Lys dans La Vallée*	1835	102,136	100,659
Honoré de Balzac	*La Cousine Bette*	1846	149,037	153,528
Honoré de Balzac	*Le Père Goriot*	1835	91,639	102,971
Honoré de Balzac	*Le Contrat de Mariage*	1835	48,142	49,043
Honoré de Balzac	*Adieu*	1830	15,484	15,277
Honoré de Balzac	*Colonel Chabert*	1832	23,138	23,908
Honoré de Balzac	*Eugénie Grandet*	1833	64,234	65,521
Honoré de Balzac	*La Femme de Trente Ans*	1842	66,487	70,305
Guy de Maupassant	*Bel-Ami*	1885	106,962	107,919
Guy de Maupassant	*Pierre et Jean*	1887	43,498	45,683
			Total SL Word Count	**Total TL Word Count**
			1,346,166	1,402,419

Table 3: Breakdown of Bilingual Corpus.

For the monolingual corpus, the intention was to remain within the Realist genre by collecting works written by the same novelists as those found in the bilingual corpus. Table 4 details the monolingual corpus. Again, all texts (only in English) were obtained through Project Gutenberg.

Author	Original Title	Year Published	# Words in TL (English)
Gustave Flaubert	*Trois Contes*	1877	24,754
Honoré de Balzac	*Le Chef d'Oeuvre Inconnu*	1831	10,637
Honoré de Balzac	*La Peau de Chagrin*	1831	96,470
Honoré de Balzac	*Sarrasine*	1830	13,119
Guy de Maupassant	*Maupassant, Contes et Nouvelles Vol.1 : Boule de Suif et Autres Histoires*	Various dates	104,572
			Total TL Mono Word Count
			249,552

Table 4: Breakdown of Mono-lingual TL Corpus.

4.1.3. Cleaning the Corpora

The bilingual corpus – which began as a collection of UTF-8 encoded text files (.txt) – required rigorous cleaning before being properly aligned and formatted as a Translation Memory Exchange file (TMX). Similarly, the monolingual corpus required some cleaning, although it would be fed to the engine in the same format (.txt files) as they were obtained. (See Appendix II.)

Firstly, each text was stripped of any unnecessary content (using Notepad++ text editing software) such as prefaces, introductions, appendices or licenses. This step insured that text pairs began and ended with corresponding sentences.

The second step of cleaning consisted in removing any unwanted fragments from the text. For example, all English texts contained page numbers which needed to be removed. Moreover, anticipating problems at the alignment process, dialogue schemes – which are unique in French – were removed. In French, the first utterance in a dialogue is wrapped within '«' and '»' (French quotation marks). Then, each change in speaker is denoted with a hyphen, '—'. For the English texts, quotation marks were removed. Short Python scripts were written to execute these two tasks, iterating through all texts. (See Appendix III for Python scripts.)

The final step of cleaning involved reformatting the files in terms of paragraphs and line breaks. Both French and English texts were in the format of a digitalized book (roughly A5 sized), so lines were consistently split by paragraph breaks even if that meant splitting sentences. Reformatting is a crucial procedure, as faulty paragraph breaks can hinder the segmentation of text which is essential for alignment as well as SMT training. For the monolingual corpus, a faulty format is also detrimental to training the language model. Figure 10 demonstrates the process on a small excerpt of text from Guy de Maupassant's *Bel Ami*.

Toutes les femmes avaient les yeux sur lui.¶
Mme Walter murmura de sa voix lente :¶
– Vous feriez avec vos souvenirs une¶
charmante série d'articles.¶
Alors Walter considéra le jeune homme pardessus le verre de ses lunettes, comme il faisait¶
pour bien voir les visages. Il regardait les plats¶
par-dessous.¶
Forestier saisit le moment :¶
– Mon cher patron, je vous ai parlé tantôt de¶
M. Georges Duroy, en vous demandant de me¶
l'adjoindre pour le service des informations¶
politiques. Depuis que Marambot nous a quittés,¶
je n'ai personne pour aller prendre des¶
renseignements urgents et confidentiels, et le¶
journal en souffre.¶

↓

Toutes les femmes avaient les yeux sur lui. Mme Walter murmura de sa voix lente : Vous feriez avec vos souvenirs une charmante série d'articles.¶
¶
Alors Walter considéra le jeune homme par- dessus le verre de ses lunettes, comme il faisait pour bien voir les visages. Il regardait les plats par-dessous.¶
¶
Forestier saisit le moment : Mon cher patron, je vous ai parlé tantôt de M. Georges Duroy, en vous demandant de me l'adjoindre pour le service des informations politiques. Depuis que Marambot nous a quittés, je n'ai personne pour aller prendre des renseignements urgents et confidentiels, et le journal en souffre.¶
¶

Fig. 10: Re-formatting of text into appropriate scheme. (Resulting format on the bottom).

4.1.4. Aligning the Bilingual Corpus

The alignment process was done using the MemoQ alignment software – with the corpus being split by novels in source-target pairs. Figure 11 shows the display of the alignment process, with the SL (French) or the left and TL (English) on the right, with links connecting corresponding segments.

16.	Et Godeschal reprit la phrase commencée : Rendue en...		And Godeschal took up the sentence he was dictating--_given in	16.
17.	Y êtes-vous ? demanda-t-il. Oui, crièrent les trois copistes.		Are you ready?{1}Yes, cried the three writers.	17.
18.	Tout marchait à la fois, la requête, la causerie et la conspiration.{1}Rendue en...		It all went all together, the appeal, the gossip, and the conspiracy.{1}Given in--	18.
19.	Hein ? papa Boucard, quelle est la date de l'ordonnance ? il faut mettre les points sur les i, saquerlotte !		Here, Daddy Boucard, what is the date of the order? We must dot our i's and cross our t's, by Jingo!	19.
20.	Cela fait des pages.		it helps to fill the pages.	20.
21.	Saquerlotte ! répéta l'un des copistes avant que Boucard le maître clerc n'eût répondu.		By Jingo! repeated one of the copying clerks before Boucard, the head clerk, could reply.	21.
22.	Comment, vous avez écrit saquerlotte ? s'écria Godeschal en regardant l'un des nouveaux venus d'un air à la fois sévère et goguenard.		What! have you written _by Jingo_? cried Godeschal, looking at one of the novices, with an expression at once stern and humorous.	22.
23.	Mais oui, dit le quatrième clerc en se penchant sur la copie de son voisin, il a écrit : Il faut mettre les points sur les i, et sakerlotte avec un k.		Why, yes, said Desroches, the fourth clerk, leaning across his neighbor's copy, he has written, '_We must dot our i's_' and spelt it by Gingo!	23.
24.	Tous les clercs partirent d'un grand éclat de rire.		All the clerks shouted with laughter.	24.
25.	Comment ! monsieur Huré, vous prenez saquerlotte pour un terme de droit, et vous dites que vous êtes de Mortagne ! s'écria Simonnin.		Why! Monsieur Hure, you take 'By Jingo' for a law term, and you say you come from Mortagne! exclaimed Simonnin.	25.
26.	Effacez bien ça ! dit le principal clerc.		Scratch it cleanly out, said the head clerk.	26.
27.	Si le juge chargé de taxer le dossier voyait des choses pareilles, il dirait qu'on se moque de la barbouillée !		If the judge, whose business it is to tax the bill, were to see such things, he would say you were laughing at the whole boiling.	27.
28.	Vous causeriez des désagréments au patron.		You would hear of it from the chief!	28.
29.	Allons, ne faites plus de ces bêtises-là, monsieur Huré !		Come, no more of this nonsense, Monsieur Hure!	29.

Fig. 11: Bilingual corpus alignment using MemoQ software.

The software employs an automatic alignment feature which connects segments by their length. For example, a one-word segment in the SL would be aligned to a one-word segment in the TL. The algorithm performs best on fewer amounts of segments (i.e. small texts), so a the longer books within the corpus were split into smaller sections. Although the algorithm was quite accurate, there would be significant portions of the texts which were improperly aligned either because the connection between an SL and TL segment was wrong or an SL segment had no link to any TL segment. For this reason, the entirety of the corpus had to be checked and properly aligned.

4.2. Choosing and Training an Engine

A SMT engine was built (using the cleaned corpora) on the KantanMT software. In addition, a series of proprietary corpora sets (KantanLibrary sets) were experimented with in order to produce the best possible engine. The relevant packaged libraries consisted of two sets of general-purpose bilingual corpora and a dictionary bilingual corpus (composed of word or phrase translations). It is important to note that the French Realist literature corpus remained the most significant within the corpus. With these three additional resources, several engines – with different combinations of the KantanLibrary corpora – were built.

At the end of the training process, the two most promising engines were manually assessed in order to determine the better of the two. Table 5 details the training data for both the engines. The monolingual corpus comprised of French Realist literature remained the same for both.

Candidate Engine 1		
Included in Bilingual Corpus	**Total Word Count**	**Unique Word Count**
19th Century French Realist Literature Corpus	≈ 5,000,000	281,680
KantanLibrary French-English General Library 1		
KantanLibrary French-English General Library 2		
KantanLibrary French-English Dictionary Library		
Candidate Engine 2		
Included in Bilingual Corpus	**Total Word Count**	**Unique Word Count**
19th Century French Realist Literature Corpus	≈ 4,000,000	213,073
KantanLibrary French-English Dictionary Library		

Table 5: Bilingual corpora of two candidate engines.

A new text was selected to test the engines, with the requirement that it should be within the French Realist genre and it should not be present in either the bilingual corpus or mono-lingual corpus. The chosen text was an excerpt from the second chapter of Stendhal's *Lucien Leuwen*. The excerpt, which totaled 1,288 words, was carefully selected to include both prose and dialogue – two principle components of realist literature.

The raw output translations of both engines were scrutinized for mistranslations, errors and other textual infelicities, using an error classification criteria primarily defined by the problems in MT described in Section 2.4. Table 6 shows the errors counted in each of the engines.

Type of Error	# Errors in Candidate Engine 1	# Errors in Candidate Engine 2
Lexical Ambiguites/Terminology: - Part of speech recognition error (word recognised as verb instead of noun) - Incorrect translation of term/word (e.g. inappropriate synonym) - Translatable text left in French - Untranslatable text translated into target language	44	61
Morphology: - Incorrect verbal form (e.g. infinitive instead of imperative) - Lack of agreement (masculine/feminine; singular plural) - Incorrect gender - Incorrect plural form	34	27
Syntax Errors and Structural Ambiguities: - Incorrect word order - Incorrect use of articles - Incorrect use of prepositions	50	50
Punctuation/Spelling: - Incorrect use of punctuation - Incorrect use of capitalization - Spelling errors	20	25
Idiomatic Elements: - Inappropriate translation of speech - Problems managing cultural or language specific references or forms - Non-coherent/inconsistent/inappropriate form of address - Idioms and colloquations incorrectly translated	4	4

Table 6: Error Classification for two candidate engines over test translation.

Overall, the raw output of both engines was – as expected – fairly poor, although the first candidate engine displayed fewer errors. The larger corpus produced a slightly more robust engine. Indeed, the second engine displayed considerably more non-translated words (left in French) in the output, accounting for the difference in errors. In terms of morphology, the difference in errors occurs because the first engine displayed slightly more gendering and pronoun errors – an inevitable issue given the language pair.

For syntax and structural ambiguities, both engines produced the same amount of errors. Indeed, where one engine's output displayed an error, the other engine displayed one too, although sometimes combined with another error category.

In terms of punctuation and spelling, the greatest source of error was incorrect use of capitalization, which counted for over 60% of the errors in this category.

Finally, both engines performed equally in terms of idiomatic elements. The raw outputs were noticeably well trained in expressions used in dialogue between characters, a feat that can be attributed to the French Realist corpus. Still, there were occasions where one engine would translate correctly whereas the other would produce an error. For example, the expression *Dieu soit loué* was properly translated by the first engine to *thank God* as opposed to the second candidate's literal translation, *God is leased*. (See Appendix IV for detailed error classification.)

Therefore, on the basis that it displayed fewer errors, the first engine – with the larger bilingual corpus – which was named the French Realist MT (FRMT), was selected to be carried forward for human evaluation.

4.3. Automatic Evaluation

KantanMT provides a streamlined process for automatic evaluation, with comprehensive details on BLEU metric (bilingual evaluation understudy), F-Measure (precision and recall ratio) and TER (Translation Error Rate).

The BLEU metric is widely used in MT evaluation which evaluates the quality of translations on a 0-1 scale (or percentage) in comparison to a reference (human) translation (Reiter, 2018).

F-measure is another measure of an MT's accuracy in relation to a reference translation. It considers precision and recall, which assess the engine's ability to retrieve the relevant candidate words for the translation (Isichei, 2018).

Finally, the TER metric measures the amount of post-editing required to adapt a translation to exactly match the target translation (Isichei, 2018). This measure is particularly useful for estimating logistics and cost of post-editing, and although that is not the goal of this project, it is nonetheless valuable.

Table 7 shows the average results for BLEU, F-Measure and TER for the FRMT provided by KantanMT Engine Report, which uses 500 segments of the bilingual corpus as testing data.

	BLEU				Avg: 47%
Interval	**<40%**	**40-49%**	**50-69%**	**70-89%**	**90-100%**
Segments	192	105	111	4	88
	F-Measure				**Avg: 49%**
Interval	**<40%**	**40-49%**	**50-69%**	**70-89%**	**90-100%**
Segments	178	46	135	48	93
	TER				**Avg: 69%**
Interval	**<40%**	**40-49%**	**50-69%**	**70-89%**	**90-100%**
Segments	121	10	121	37	211

Table 7: Results of automatic evaluation metrics for FRMT.

4.4. Designing a Human Evaluation Survey

Although the automatic evaluation metrics provide a solid idea of the quality of the FRMT, the nuanced qualities of literature are best evaluated by potential human readers. As per the previous experiments in MT for literature (see Section 3.2.1.), for a field that involves just as much comprehension as it does human interpretation and appreciation, the most significant method of assessment is through human evaluation (Moorkens et. al., 2018; Besacier, 2014).

A total of 12 readers – a mixture of bilingual (English and French) as well as monolingual English speakers – were surveyed on the quality of the French Realist MT output. The source text that was translated was an excerpt of Guy de Maupassant's *Une Vie* (1883), another text from the Realist domain. The text selected was in line (in terms of form and complexity) with respect to the Realist corpus used for the engine.

In an effort to avoid prior bias in relation to the FRMT engine's literary purpose, the engine was named 'CustomMT' for the survey. (See Appendix V.)

5. Human Evaluation

5.1. Survey Information

Along with quality evaluation of the FRMT output and specific comprehension questions, readers were also asked to compare the FRMT output with the quality output by GT – a general purpose SMT with billions of training data from a variety of sources. The goal with this comparison was to determine whether the FRMT displayed any specific or characteristic traits given its literary corpus and training.

The evaluation was output-only, where evaluators are not given the original text. The objective in this type of evaluation is to focus solely on the (literary) quality of the output text (Sommers, 2003, p.214). (See Appendix V for full survey and both MT outputs.)

5.2. General Quality of Output

Figure 12 shows readers' general responses to the FRMT output text regarding readability and overall understanding of the text.

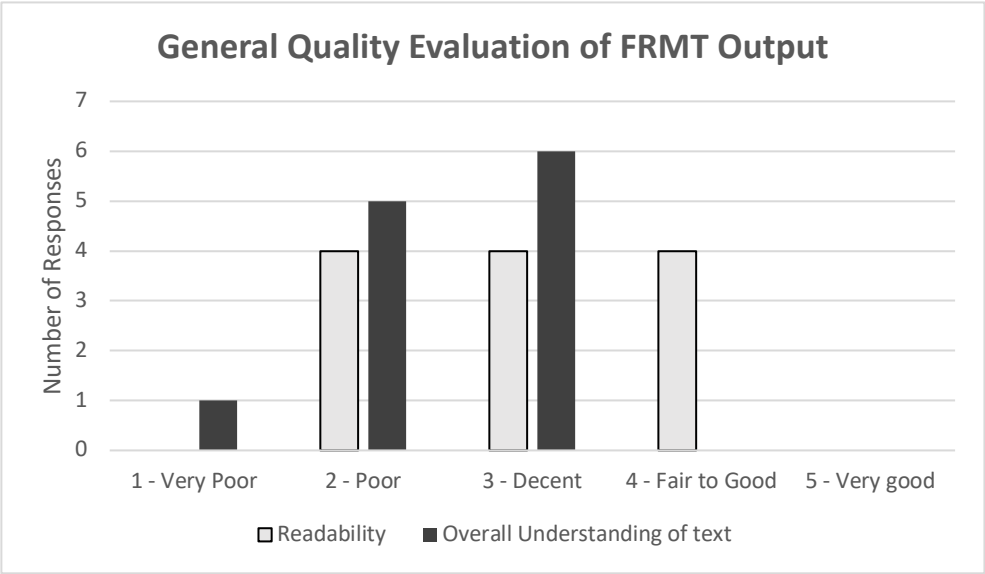

Fig. 12: Responses given on text readability and understanding.

The separation in readability is arguably accounted by some readers not comprehending French (as the text contained several non-translated words) as well as differing aptitudes towards reading. In terms of general understanding, half of the pool considered it of poor quality, with nobody assessing it as anything more than decent.

5.3. Comprehension Questions

The three straightforward comprehension questions were selected on the basis of their importance towards a satisfactory understanding of the text. Moreover, the context in which the answers lay displayed different output challenges for the MT. For example, the answer to the third question is found within a dialogue between characters, which is traditionally an instance of difficulty for MT.

Precisely half of the pool of readers answered all three questions correctly, with the other half getting one out of three wrong. In this sense, and considering the simplicity of the questions, it is clear that the FRMT was not sufficiently successful in transferring the direct meaning of text.

5.4. Error Breakdown

Readers were asked to use the Error Classification table (see Table 6 in Section 4.2.) to rank the FRMT's performance in terms of error category. Figure 13 shows the category of errors which readers deemed to be most prominent within the output translation.

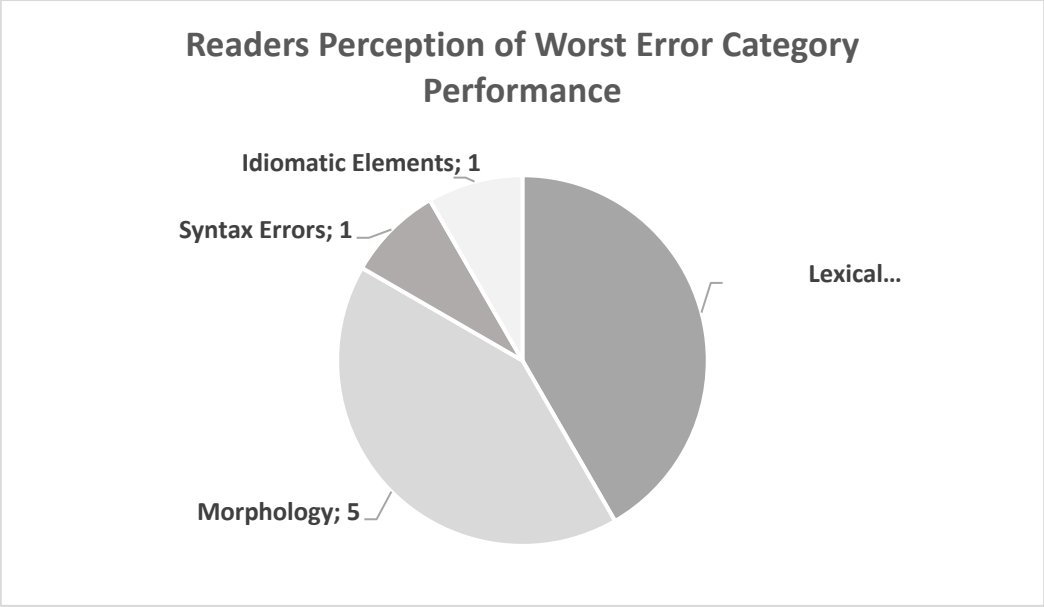

Fig. 13: FRMT output worst category of error.

For all but one reader, lexical ambiguities consistently featured in the top two worst error categories. This is due in the most part to the FRMT output involving numerous un-translated words. Moreover, there were words that had noticeably been mistranslated. Morphology was the other consistent presence, with 8 out of 12 placing it in top two worse categories. The reason for this is the FRMT's mistakes with genders.

The short extract involved both genders – a female protagonist and the protagonist's father. In numerous cases, pronouns *his* and *him* were associated to the female character. This occurred potentially because the corpus of the FRMT was male-dominated, and the French words *sa* and *lui* (which relate both female and male) contained a stronger bias towards the male translation *his* and *him*. Syntax errors, which were generally ranked close to morphology, reflect – among other mistakes – the imperfect use of prepositions. Table 8 shows an example of a sentence with the worst performance error categories according to readers.

French Realist MT Output			
his father hésitât from if the time did not éclairissait			
Source of Error	**Error Category**	**Error Subcategory**	**Correct Output**
hésitat, éclairissait	Lexical Ambiguity	Non-translated words	*hesitated, clear*
time	Lexical Ambiguity	Incorrect translation	*weather*
his	Morphology	Incorrect gender	*her*
from	Syntax Error/Lexical Ambiguity	Preposition error/ wrong part of speech	e.g. *to leave* (myriad options)

Table 8: Error classification for a faulty sentence in FRMT output.

It is also important to note that punctuation and spelling ranked, for all but one reader, as the best error category performance. Moreover, idiomatic elements tended to score favorably. Finally, the FRMT engine had clear trouble in the most significant aspects of properly formulated language, which substantially hinder the readability and comprehensibility of the text. Notwithstanding, readers seem to have considered parts of the FRMT's handling of language – namely idiomatic elements – less problematic.

5.5. Comparison with Google Translate

Readers were asked to rate the quality of the GT output (of the same extract) also by ranking the performance in terms of error category. It is important to note that the performance of each category is relative and not absolute. The GT output contained less mistakes, meaning that even the worst performing category of error was not necessarily as poor as the worst category for the FRMT. Nevertheless, the rankings demonstrate the reader's perception of what are the most prominent and troublesome kind of errors for the particular engine.
Figure 14 shows the results for both the FRMT and GT.

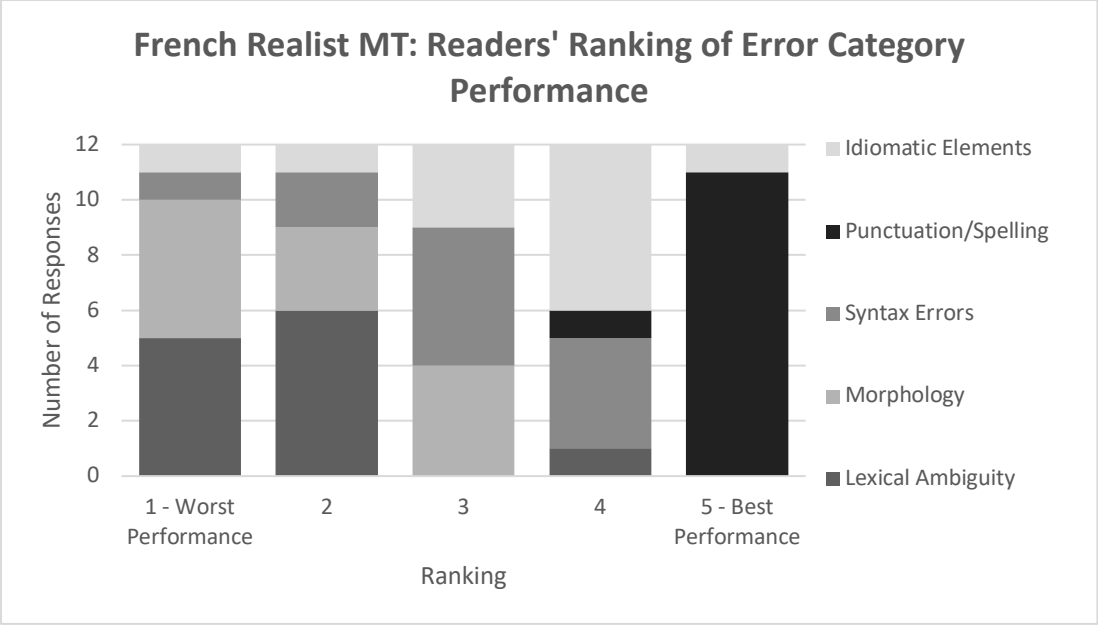

Fig. 14(a): Reader's assessment of FRMT output quality.

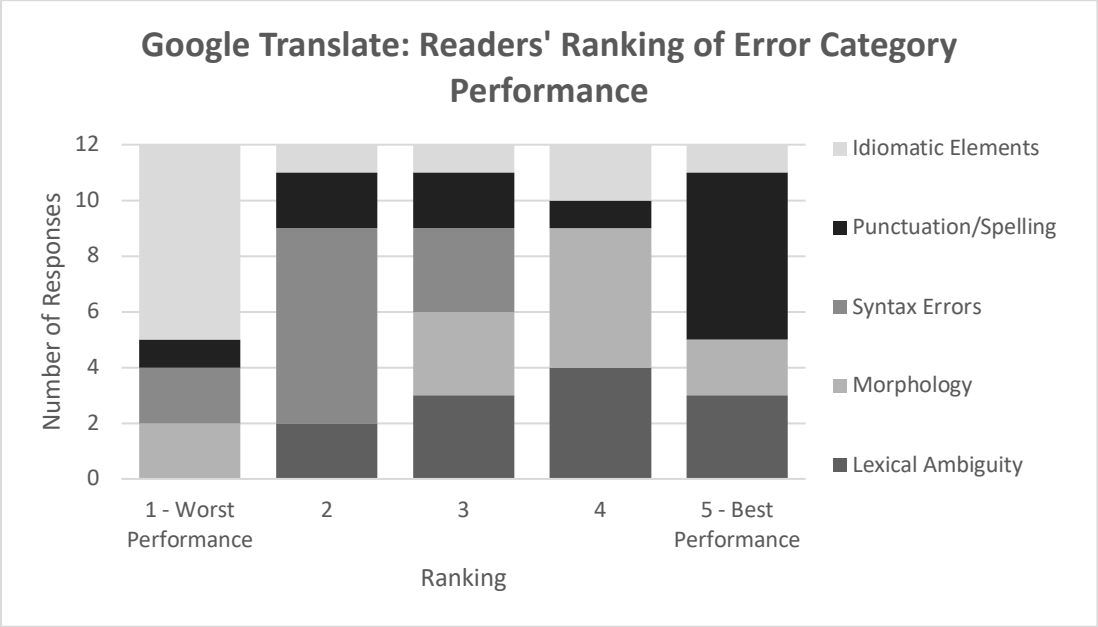

Fig. 14(b): Reader's assessment of Google Translate output quality.

The readers' assessment of GT output quality suggests that it is more fluent in writing than the FRMT. Indeed, lexical ambiguity and morphology tended to rank 4th or 5th (i.e. best) whereas for the FRMT these categories were heavily rated as having the worst performance. (Although not error free, GT has been deemed by Groves and Mundt (2015) as comparable to the minimum quality of language needed for university admission). On the other hand,

idiomatic elements prove to become more of an issue in Google Translate – as the majority of readers classified it as the worst performing category. From the two previous point, there is an argument that despite GT's near-fluency, the more nuanced aspects which are essential to the appreciation of literature – particularly concerning idiomatic elements – begin to surface. For the FRMT, although idiomatic elements may have been overshadowed by greater errors in other categories, readers assessed the performance more favorably, ranking it between 3 and 5.

5.6. Literary Appreciation

The final section of the survey questioned the pool of readers about which MT output read more like a 19th Century piece of literature, as well as asked for examples of literary appreciation from either of the engines. This was the most important test for the FRMT, to determine whether it is – having been trained on French Realist literature – more apt or capable in translating texts from the specified genre.

Overall, nine out of twelve readers answered that the FRMT was a more faithful emulation of 19th Century literature. However, this was mixed with some caveats, such as "ignoring the amount of errors" or "when readable". Regardless, the fact that three quarters of the pool surveyed believed that the FRMT displayed promising literary features is testimony to the rationale of training on such a specific literary corpus. Table 9 shows a few of the instances which readers considered more literary (for a 19th century work) in the FRMT output.

French Realist MT	**Google Translate**
the rain never ceased	*the rain did not stop*
The storm, all night, rang against the windowpanes, and the roofs.	*The shower all night had struck against the tiles and the roofs.*
she speculated on the horizon	*she had been questioning the horizon*
'if you contrive to decide to your mother, I am quite willing.'	*'if you can decide your mother, I don't mind.'*
'let us go'	*'let's go'*
And the rain, falling without respite from the previous evening, was the first great grief of his life.	*And the rain, falling relentlessly since the previous evening, was the first major sorrow in his existence.*

Table 9: Examples given by readers of literary instances in FRMT

These examples demonstrate that the FRMT often translates into more archaic words or expressions (e.g. *never ceased* vs *did not stop*) which are more related to 19th Century realist literature. Arguably, the FRMT successfully emulates the translation decisions which are present in it's training data. For example, the translations of dialogue between high-society characters which is a staple in Realist literature, seems more appropriate in the FRMT. The simple phrase *let us go* in the FRMT is a small but noticeable sign of aristocratic speech compared to GT's *let's go*, which is contemporary and does not carry hints of nobility. The same applies for *'I am quite willing'* compared to *'I don't mind'*, which portray the character of the father (an aristocrat) in a different light. In this sense the FRMT, besides privileging the equivalence (see Section 3.1) of the translation, provides a more coherent cultural adaptation – mirroring the translation choices existing within the training corpus.

The remaining readers which chose GT as the most literary translation on the basis that it was more fluent than the FRMT – conceding that although the FRMT did contain more literary instances, these were rendered irrelevant by the output as a whole which often lacked fluency. In other words, it was difficult to identify a voice or style given the chaos of the imperfections in translation. The examples given to support GT's greater literary value are longer segments, highlighting the GT's capacity to maintain cohesiveness in longer articulations.

The results of this section of the survey suggest that there is a correlation in MT between the quality of translation with regards to a specific genre and the MT's ability to perform fluently and without mistakes. Whereas FRMT falters in linguistic fluency, it performs (albeit sporadically) in the literary components that are important to translators. GT, on the other hand, performs significantly better in fluency but appears to sacrifice the literary relevance in terms of transferring culture and nuanced literary and stylistic features.

6. Conclusions and Future Recommendations

With the recent advances in SMT (and NMT), the future of MT for literature is certainly hopeful. The experiment with the FRMT shows that – although the general fluency of the engine was not ideal – it is possible to train on the translation patterns of a specific genre and obtain an engine that is versed in the given genre. Still, corpus management poses some issues; firstly, distinctions between genres and literary movements are fluid, so deciding which work falls within which genre is a challenge. Secondly, current SMT (and NMT) require a significantly sizeable corpus, meaning that a corpus of a literary genre must attain a given size which may not be logistically possible in terms of the genre definition or the availability of texts and their translations (for example more recent works are protected by copyright and licensing).

It is evident that MT – at least as a FAHQMT – is still far from reaching a level that is suitable for literature translation. Taking an MT output as a draft or a final translation of a literary piece comes with certain risks and compromises. Fundamentally, literature translation is entirely dependent on interpretation and the skill of reproducing an effect and feeling. Given each piece of literature is unique in its own right, the nuanced decision-making process entrusted to an MT does not merely entail pattern recognition and reproduction, and therefore poses translation, literary, and even philosophical problems. Indeed, even in the moments that the FRMT engine – designed for a specific literary genre – displayed literary signs and appropriate translation patterns, it is difficult to determine whether features of the source text are preserved through translation or are newly generated by the FRMT engine.

Notwithstanding, as seen from studies such as Moorkens et. al. (2018), which integrated MT into human literature translators' workflow, there is a growing idea that MT can in some capacity assist the process of literature translation. This would provide several benefits for publishers and readers alike by increasing the efficiency of translation as well as facilitate translations into different target languages. In this ambition, MT should not be regarded as a threat to human translators, but as a tool which can support a field defined by sensitivity and nuance.

Certainly, the application of MT in literature is a venture which requires the co-operation of all the constituent specialists; from computer scientists and computational linguists to linguists and literature translators. After all, the decisions and advancements in MT will invariably be human.

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Appendices

Appendix I: KantanMT Platform Screenshots

Figures 1, 2 and 3 show the key features of the KantanMT platform used for the project.

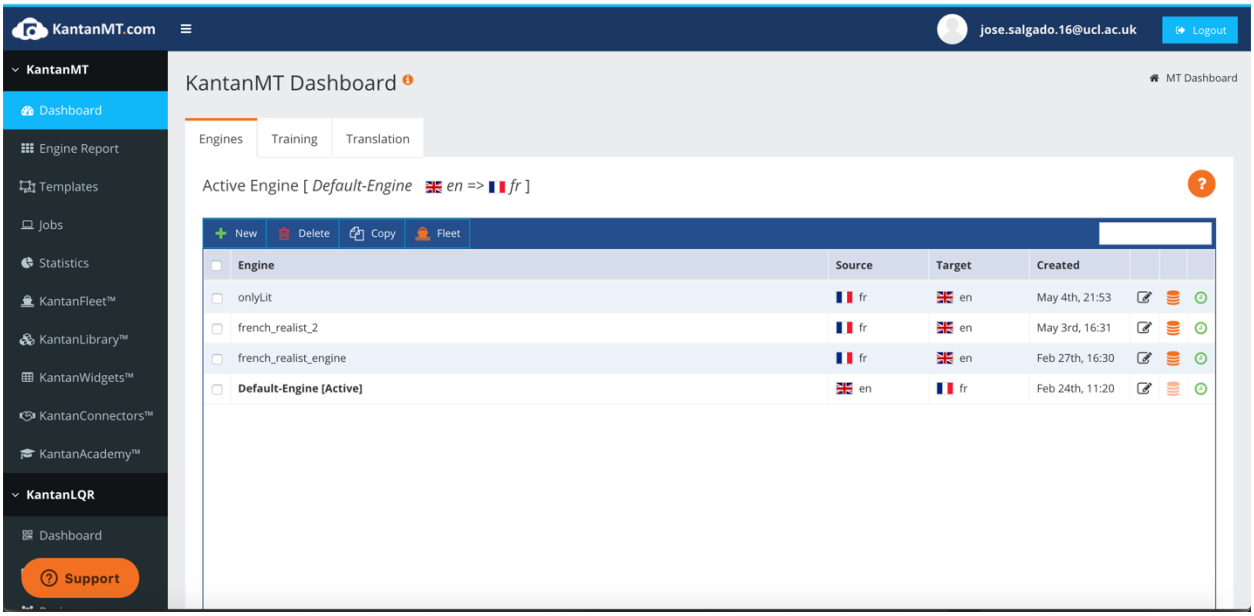

Fig. 1: KantanMT Dashboard with Engine, Training and Translation tabs.

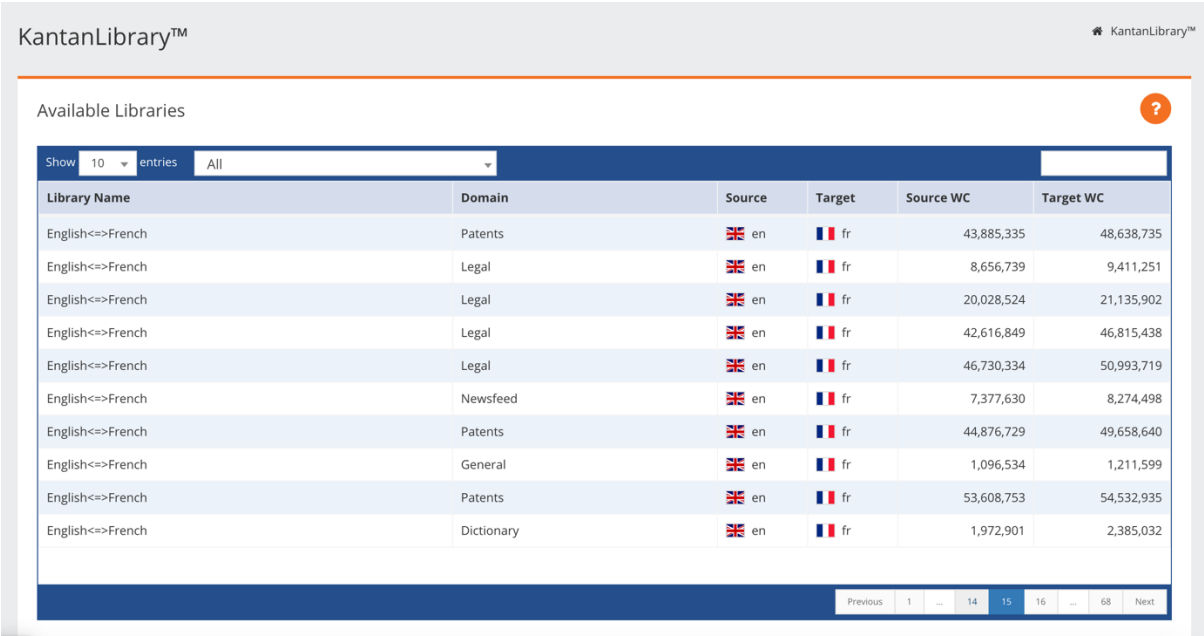

Fig. 2: KantanLibraries feature with example EN-FR (bi-directional) libraries.

Engine	Source	Target	WC	TWC	MWC	UWC	F-Measure	BLEU	TER
Default-Engine	en	fr	0	0	0	0	0%	0%	0%
french_realist_2	fr	en	4,728,897	1,265	141,986	281,680	49%	46%	72%
french_realist_engine	fr	en	3,510,732	23,018	141,986	213,073	49%	47%	69%
onlyLit	fr	en	1,231,192	0	0	79,208	48%	21%	71%

Fig. 3: Engines Performance Report. The orange stack symbol on the right links to each engine's detailed Analytics report.

Appendix II: Bilingual and monolingual corpora texts

Links to access all texts found within the bi-lingual (Table 1) and monolingual corpus (Table 2). All clean versions can be found at https://github.com/josesalgado97/French-Realist-MT-Experiment.

Author	Title	Original Text (Link)	English Translation (Link)
Stendhal	*Le Rouge et Le Noir*	https://beq.ebooksgratuits.com/vents/Stendhal-rouge.pdf	https://www.gutenberg.org/ebooks/44747
Stendhal	*La Charteuse de Parme*	https://beq.ebooksgratuits.com/vents/Stendhal-chartreuse.pdf	https://www.gutenberg.org/ebooks/57638
Gustave Flaubert	*Madame Bovary*	https://www.beq.ebooksgratuits.com/vents/Flaubert-Bovary.pdf	https://www.gutenberg.org/ebooks/2413
Gustave Flaubert	*Education Sentimentale*	https://beq.ebooksgratuits.com/vents/Flaubert-education.pdf	https://www.gutenberg.org/ebooks/27537
Honoré de Balzac	*Le Lys dans La Vallée*	https://beq.ebooksgratuits.com/balzac/Balzac-34.pdf	https://www.gutenberg.org/ebooks/1569
Honoré de Balzac	*La Cousine Bette*	https://beq.ebooksgratuits.com/balzac/Balzac-55.pdf	https://www.gutenberg.org/ebooks/1749
Honoré de Balzac	*Le Père Goriot*	https://beq.ebooksgratuits.com/balzac/Balzac-39.pdf	https://www.gutenberg.org/ebooks/1237
Honoré de Balzac	*Le Contrat de Mariage*	https://beq.ebooksgratuits.com/balzac/Balzac-19.pdf	https://www.gutenberg.org/ebooks/1556
Honoré de Balzac	*Adieu*	https://beq.ebooksgratuits.com/balzac/Balzac_76_Adieu.pdf	https://www.gutenberg.org/ebooks/1554
Honoré de Balzac	*Colonel Chabert*	https://beq.ebooksgratuits.com/balzac/Balzac_40_Le_colonel_Chabert.pdf	https://www.gutenberg.org/ebooks/1954
Honoré de Balzac	*Eugénie Grandet*	https://beq.ebooksgratuits.com/balzac/Balzac-26.pdf	https://www.gutenberg.org/ebooks/1715
Honoré de Balzac	*La Femme de Trente Ans*	https://beq.ebooksgratuits.com/balzac/Balzac-18.pdf	https://www.gutenberg.org/ebooks/1950

Guy de Maupassant	*Bel-Ami*	https://beq.ebooksgratuits.com/vents/Maupassant_Bel_Ami.pdf	https://www.gutenberg.org/ebooks/33928
Guy de Maupassant	*Pierre et Jean*	https://beq.ebooksgratuits.com/vents/Maupassant_Pierre_et_Jean.pdf	https://www.gutenberg.org/ebooks/3804

Table 1: Bi-Lingual Corpus

Author	**Original Title**	**English Translation (Link)**
Gustave Flaubert	*Trois Contes*	https://www.gutenberg.org/ebooks/10458
Honoré de Balzac	*Chef d'Oeuvre Inconnu*	https://www.gutenberg.org/ebooks/23060
Honoré de Balzac	*Peau de Chagrin*	https://www.gutenberg.org/ebooks/1307
Honoré de Balzac	*Sarrasine*	https://www.gutenberg.org/ebooks/1715
Guy de Maupassant	*Maupassant, Contes et Nouvelles Vol.1 : Boule de Suif et Autres Histoires*	https://www.gutenberg.org/ebooks/21327

Table 2. Monolingual Corpus

Appendix III: Data Cleaning Python Scripts

Figure 4 shows the script written to clean all French texts. It performs basic functions on an input file and writes to an output file. SL (French) are iterated over, one text at a time. Using the '.replace()' function, the dialogue characters ('«' , '»' , and '–') are cleared from each line of the text. Finally, if the line is in integer form (i.e. if it is a page number) then it is not written to the output file. Figure 5 shows the cleaning script for all English texts. It is equal to that of the French, except that the dialogue characters to remove are the traditional quotation marks. Both scripts are available at https://github.com/josesalgado97/French-Realist-MT-Experiment.

```
import os
import glob

path = "/Users/zeca/Desktop/corpus_cleaning/texts_french/*.txt"

for file in glob.glob(path):            # iterate over all files
    with open(file, "r") as input_file:
        filename = os.path.basename(file)   # get input file name
        output = ('%s_clean.txt' % filename)     # meaningfully name the output file
        with open(output, "w") as output:
            for line in input_file:
                line = line.replace('–', '')
                line = line.replace('«', '')    # replace problematic characters
                line = line.replace('»', '')
                line = line.strip()     # clear white space
                try:
                    int(line)  # check if line is a number
                except ValueError:
                    output.write(line + "\n")
```

Fig. 4: Cleaning script for SL (French) texts

```
import os
import glob

path = "/Users/zeca/Desktop/corpus_cleaning/texts_english/*.txt"

for file in glob.glob(path):            # iterate over all files
    with open(file, "r") as input_file:
        filename = os.path.basename(file)   # get input file name
        output = ('%s_clean.txt' % filename)     # meaningfully name the output file
        with open(output, "w") as output:
            for line in input_file:
                line = line.replace('“', '')
                line = line.replace('”', '')
                line = line.strip()     # clear white space
                try:
                    int(line)  # check if line is a number
                except ValueError:
                    output.write(line + "\n")
```

Fig. 5: Cleaning script for TL (English) Texts

Appendix IV: Detailed Error Classification for Candidate Engines

Tables 3 and 4 give a detailed error analysis of the two candidate engines (on an extract of Stendhal's *Lucien Leuwen*) that were tested to determine highest quality MT engine.

Source Text	Raw FRMT Output	Main Error Category	Error Subcategory
lancier	*lancier*	Lexical Ambiguity/Terminology	non-translated word
alors comme alors	*then as well*	Idiomatic Element	Idiom incorrectly translated
première fois de sa vie	*first time in her life*	Morphology	Incorrect Gender
les sentinelles des Tuileries lui portèrent les armes	*the sentinels Tuileries portèrent her arms*	Lexical Ambiguity/Terminology, Morphology	word order, untranslated word, incorrect gender
Après dix minutes, qui lui parurent un siècle	*After ten minutes, who seemed te her a century*	Syntax Error, Morphology	preposition, gender
Grand Dieu! Est-ce là un héros? S'écria-t-il enfin	*great God! Is this a hero If écria he at last*	Punctuation, punctuation, syntax, Lexical ambiguity	incorrect capitalization, incorrect punctuation, preposition, non-translated word
Que veut dire cette mine de dégout	*that means that expression of disgust*	Syntax Error	preposition
Veux-tu ou ne veux-tu pas	*You want, or you don't want*	Syntax Error	preposition missing
Ton père te donne de quoi vivre	*Your father will give you what live*	Morphology, Syntax Error	verb tense, preposition missing
Vil ou non, il t'est mille fois supérieur	*sell or not, it is you a thousand times higher*	Punctuation, Syntax, Lexical ambiguity	incorrect capitalization, incorrect gender/pronoun, incorrect translation
Tu as cent fois raison	*You have a hundred times right*	Morphology, Lexical Ambiguity	incorrect verb form, incorrect translation
J'ai horreur de la porte par laquelle il faut passer	*I have a horror of the door by which he must pass*	Syntax Error, Morphology	incorrect articles, incorrect preposition, gender/pronoun
Que serai-je sans vous?	*That would I without you?*	Syntax Error, Morphology	preposition error, missing verb
Dieu soit loué!	*thank God*	Punctuation	incorrect capitalization
Ce qu'il voyait le choquait	*What he saw, on the*	Lexical Ambiguity, Syntax Error	incorrect word order, incorrect translation

Table 3: Error Classification for Candidate Engine 1

Source Text	Raw FRMT Output	Main Error Category	Error Subcategory
lancier	*lancier*	Lexical Ambiguity	non-translated word
alors comme alors	*then as then*	Idiomatic Element	Idiom incorrectly translated
première fois de sa vie	*first time in her life*	Morphology	Incorrect Gender
les sentinelles des Tuileries lui portèrent les armes	*the sentinels Tuileries portèrent her arms*	Lexical Ambiguity/Terminology, Morphology	word order, untranslated word, incorrect gender
Après dix minutes, qui lui parurent un siècle	*After ten minutes, who appeared a century*	Syntax Error, Morphology	preposition error
Grand Dieu! Est-ce là un héros? S'écria-t-il enfin	*great God! Is it là a héros if écria he at last*	Punctuation, Syntax Error, Lexical ambiguity	incorrect capitalization, incorrect punctuation, preposition, non-translated word
Que veut dire cette mine de dégout	*What was the meaning of expression of disgust*	Morphology	incorrect verb form
Veux-tu ou ne veux-tu pas	*You want, or you don't want*	Syntax Error	preposition missing
Ton père te donne de quoi vivre	*Your father will give you what live*	Morphology, Syntax Error	verb tense, preposition missing
Vil ou non, il t'est mille fois supérieur	*sell or no, he is a thousand times you higher*	Punctuation, Syntax, Lexical Ambiguity, Morphology	incorrect capitalization, incorrect translation, incorrect word order, missing preposition
Tu as cent fois raison	*You have a hundred times right*	Morphology, Lexical Ambiguity	incorrect verb form, incorrect translation
J'ai horreur de la porte par laquelle il faut passer	*I have a horror of the door by which we must pass*	Syntax Error, Morphology	incorrect articles, incorrect preposition
Que serai-je sans vous?	*What should I be without you?*	Morphology	incorrect auxilary verb
Dieu soit loué!	*God is leased*	Punctuation, Lexical Ambiguity	incorrect translation, missing punctuation
Ce qu'il voyait le choquait	*That he saw the airs of independence*	Lexical Ambiguity, Syntax Error	incorrect article, incorrect translation

Table 4: Error Classification for Candidate Engine 2

Appendix V: Human Evaluation Survey & FRMT and GT Output

Below is the human evaluation survey and FRMT (named CustomMT for the survey) and GT outputs as they were sent to the pool of readers.

MT Human Evaluation Survey

Notice: By partaking in this survey, you agree that the feedback you share can be used for the purposes of student dissertation research on MT output quality. All responses are anonymous and require no personal information. For any questions or concerns, contact the student researcher at jose.salgado.16@ucl.ac.uk.

Instructions

- You have received two different Machine Translations of the same text – an extract from Guy de Maupassant's *Une Vie* (1883). The first (Custom MT Output) is a customized engine built for this project. The second (GoogleTranslate Output) is the translation provided by Google Translate. Please start by reading the Custom MT output.

- Please view the Error Category table before reading the text, and take note (mental note is fine) of any errors that you notice in the text, you will need it for Question 3 of Part 1.

Error Category Table

Lexical ambiguity (a) : wrong translation of words, non-translated words (i.e. left in French), wrong part of speech (e.g. noun instead of adjective)
Morphology (b): Incorrect verb form, Incorrect gender, incorrect plural
Syntax Errors (c): incorrect word order, preposition errors (e.g. of, to, from), incorrect articles (e.g. the, an, a)
Punctuation/Spelling (d): incorrect punctuation, spelling errors, capitalisation errors
Idiomatic elements (e): inappropriate translation of speech, problems managing cultural or language specific collocations or references

- Once you've finished reading the Custom MT Output, respond to Part 1 and 2 below.

- Next, read the Google Translate Output – also making notice of errors along the way – and respond to the questions in Part 3.

- Thank you!

Part 1: Output Quality Questions

1) Is the text readable? 1 - 5 (not at all readable - very good)

Answer :

2) Is the content understandable ? 1 – 5 (Not at all – very clear & understandable)

Answer :

3) Rank by error category. (reference the Error Category Table):

Ranking of Error Categories – Worst (1) to Best Performance: 1)
2)
3)
4)
5)

Part 2: Comprehension Questions

1) Where is Jeanne departing from?

Answer:

2) What has Jeanne forgotten?

Answer:

3) What must Jeanne do in order to convince her father to depart?

Answer:

Part 3: Compare to Google Translate Output

1) Rank the output quality with respect to the same Error Categories. (Reference the Error Category Table on previous page.)

Ranking of Error Categories – Worst to Best Performance: 1)
2)
3)
4)
5)

2) Which translation seems more literary / related to 19th Century?

Give a 1-2 examples found in the output (word, phrase, etc.)

Answer :

CustomMT Output

Jeanne, Having finished his trunks, went up to the window, but the rain never ceased.

The storm, all night, rang against the windowpanes, and the roofs. The sky, and with low water, seemed flat, vidant on earth, the délayant slurry, like the bursting of sugar. Manage passed, full of a heavy heat. The rumbling of the streams inundated filled streets désertes where homes, like a sponge, drank the water, which was pénétrait audedans and sweat the walls of the cellar to the attic.
Jeanne, out on the eve of the convent, free at last for ever, ready to all the joys of life which dreamlike so long, feared that his father hésitât from if the time did not éclaircissait, and for the hundredth time since the morning she speculated on the horizon.

Then, she aperçut that she had forgotten to her calendar in his travelling bag. She cueillit on the wall, the little board divided a month, and wearing the middle of a drawing the date of current year, 1819, in figures d'or. Then, she biffa pencil-strokes of the four first columns, rayant every name of saint up on 2 May, day of his exit from the convent.

She ran to his father and kissed her, in the étreignant :
"well, are we going? " she said.
He smiled, shook her hair already white and that he was quite long and, holding out his hand towards the window :
"How do you travel by such a time? " but she was praying, câline and tender "Oh, papa, let us go, I beg of you. It will be good in the afternoon.
- But your mother only consentira ever.
- If, I promise you, I live.
- If you contrive to decide to your mother, I am quite willing. " and she précipita to the room of the Baroness. For she had expected date of departure with growing impatience.

Since his entry to the Sacred Heart, she had not left Rouen, his father only for any amusement before the age which he had set in. Twice only had been emmenée fifteen days Paris, but it was a city again, and she did not dreamlike that the campaign. Now that she was going to spend the summer in their property of peoples, old castle of family planted on the cliff near Yport. And she promised himself a joy infinite of Free Life on the edge of the water. Then he heard that they put him to donation of the manor, that she always habiterait when she would be married. And the rain, falling without respite from the previous evening, was the first great grief of his life.

Google Translate Output

Jeanne, having finished her trunks, approached the window, but the rain did not stop.
The shower all night had struck against the tiles and the roofs. The sky, low and charged with water, seemed to be burst, emptying itself on the ground, dissolving it in mush, melting it like sugar. Gusts passed, full of heavy heat. The hum of overflowing streams filled the deserted streets where the houses, like sponges, drank the humidity which penetrated inside and made the walls of the cellar sweat in the attic.
Jeanne, who had left the convent the day before, finally free forever, ready to seize all the joys of life of which she had dreamed for so long, feared that her father would hesitate to leave if the weather did not clear, and for the hundredth time since morning she had been questioning the horizon.

Then she realized that she had forgotten to put her calendar in her travel bag. She picked up on the wall the small cardboard divided by month, and bearing in the middle of a drawing the date of the current year, 1819, in gold figures. Then she crossed out the first four columns with a pencil, crossing out each saint's name until May 2, the day she left the convent.

She ran to her father and kissed him, hugging him, "Well, are we going? " she says. He smiled, shook his already white hair, which he wore long enough, and, reaching out for the window, "How do you want to travel in such weather?" But she begged him, cuddly and tender: "Oh! Dad, let's go, I beg you. The weather will be fine in the afternoon.
- But your mother will never agree.
- Yes, I promise you, I'll do it.
- If you can decide your mother, I don't mind. And she rushed to the baroness's room. Because she had waited for this day of departure with growing impatience.

Since entering the Sacred Heart she had not left Rouen, her father allowing no distraction before the age he had set. Only twice had she been taken a fortnight to Paris, but it was still a city, and she only dreamed of the countryside. She was now going to spend the summer on their Peoples property, an old family castle on the cliff near Yport; and she promised herself an infinite joy of this free life on the edge of the waves. Then it was understood that she was given this mansion, which she would always live in when she was married. And the rain, falling relentlessly since the previous evening, was the first major sorrow in his existence.